Functions for Leave-One-Out Validation in GLMs

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Cross-validation: evaluating predictions on held-out data

What happens when we use the same data twice?

$$\widehat{\beta} = \text{fit}(D), \qquad \text{error} = \text{test}(D, \widehat{\beta}).$$

- The fitted coefficients adapt to both signal and random variation in D.
- Testing on that same data usually gives optimistic error: predictions look better than they would for new subjects.

A basic solution: separate training and testing data

Without an external dataset, randomly split the available subjects 2:1.

Training set: about $2/3$	**Testing set: about** $1/3$
Estimate the model coefficients.	Predict outcomes using the training coefficients, calculate MAE and RMSE without refitting.

Target: prediction error for a new subject from the same population.

Cross-validation approaches

Approach	**How it uses limited data**	**Key feature**
Holdout split	Fit on a training set, evaluate on a separate test set.	Results depend on the chosen split, less training data can reduce power.
K-fold CV	Split into K roughly equal folds, train on $K-1$ and test on the remaining fold each round.	Each subject is held out and tested exactly once.
LOO CV	Use $K = n$: hold out one subject per round.	Keeps $n - 1$ subjects for training.

LOO is useful when data are limited.
LOO requires one fit per subject and can be computationally expensive.

LOO example: five subjects, five fits

Round	Fit coefficients using subjects	Held-out subject
1	2, 3, 4, 5	1
2	1, 3, 4, 5	2
3	1, 2, 4, 5	3
4	1, 2, 3, 5	4
5	1, 2, 3, 4	5

$$\eta_i = x_i^{\mathsf{T}} \widehat{\beta}_i, \qquad \widehat{y}_i = g^{-1}(\eta_i), \qquad i = 1, \ldots, 5.$$

x_i: held-out subject i's predictors, including the intercept.
Re-estimate $\widehat{\beta}_i$ each round, the inverse link gives the response-scale prediction $\widehat{y}_i$.

- Train on $n - 1$ subjects. The held-out subject does not enter that fit.
- Collect n predictions, one for each subject.

This project: 50 subjects, 50 independent fits, 49 training subjects per fit.

Summarizing LOO prediction errors

For each held-out subject:

$$\text{error} = \text{predicted} - \text{observed}.$$

Mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\text{error}_i|$$

- Easy to interpret.
- Less affected by extreme errors than RMSE.

Root mean square error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \text{error}_i^2}$$

- Gives greater weight to large errors.

Both use the outcome's units. Lower is better on the same evaluation data.
Different outcome types are supported, binary predictions are probabilities.